

Course III: Corporate Finance

Summer Term 2026/25

Unit 4 Mini-Cases

Please upload your solutions and indicate the Mini-Cases you have solved before 10:00 p.m. on April 12, 2026. There will be no deadline extensions!

To upload your solutions and make indications, please go to the assignment section on Canvas and choose “Unit 1 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). Only complete submissions containing answers to the T/F questions and a solutions upload will be considered. Your last complete attempt counts.

You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solution file. If your solution consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Mini Case 4.1 An all-equity-financed company has assets that will generate one of two possible cash flows next year: €400 with 30% probability, or €0 with 70% probability next year. The cash flows are uncorrelated with returns on a broad stock market index. The assets are under the supervision of a manager. If the manager exerts effort, the probability that the cash flow is €400 increases to 80%. However, exerting effort means that the manager must work in her leisure time which would result in a personal cost to her of €30. The company’s compensation committee is considering linking the manager’s pay to the realized cash flows. More specifically, the manager will receive a bonus of $x\%$ of the cash flows. What bonus should the compensation committee choose when it is trying to maximize the residual payoff to equity holders, i.e., the payoff after paying the manager’s bonus? All agents are risk neutral.

Mini Case 4.2 The company Golden Goose invests into a project that will either have a free cash flow of EUR 100,000 or EUR 50,000. The probability of the good realisation depends on the effort of the project manager. If she exerts a lot of effort, the project will generate a cash flow of EUR 100,000 with a probability of 60%. If she does not put in high effort, the probability of success is only 40%.

The manager decides whether to put effort in a project based on her expected utility (hint: the manager chooses to exert the effort if this maximizes her expected utility). The manager’s utility function is:

$$U(\text{compensation}) = \log_{10}(\text{compensation})$$

High effort reduces the manager’s utility by 0.1.

Equity investors are risk-neutral and pay the manager out of the cash flow from the project. The manager receives a fixed wage of EUR 4,000 (independent of the outcome of the project). Assume that the cost of equity capital is 0%. (Note: Use LOG10() to calculate the utility if you are using excel.)

- a. What is the expected value of equity, if the manager exerts *low* effort?
- b. What is the expected value of equity, if the manager exerts *high* effort?
- c. Will the manager choose to exert high or low effort?

Equity holders are thinking about a compensation schedule for the manager that depends on the success of the project. The manager gets 20% of the project's value. There is no fixed wage.

- d. In this case, will the manager choose to exert high or low effort?
- e. What is the expected value of equity, if the manager exerts *high* effort?
- f. Which type of compensation will the equity investors offer the manager?

Instead of paying the manager based on the project's value, the equity holders can incentivize the manager by giving him a fixed amount of bonus in addition to the fixed salary if the project succeeds:

- g. What is the minimum bonus that makes the manager indifferent between exerting high or low effort?

Mini Case 4.3 Shareholders of an all-equity financed company "Aerospace 27" expect the market value of the company at the end of 2028 to be €900 million, €600 million or €300 million with equal probability. The manager of the company, who acts in the interest of shareholders, wants to restructure the teams in order to improve productivity. However, this change is risky, since the manager does not know whether new teams will work well together. She expects the teams to work either really well or very poorly together and estimates payoff probabilities of the project with new teams to change to 40%, 5% and 55% respectively.

- a. Calculate the expected value of company's assets at the end of 2028 if the manager makes no changes in teams.
- b. Repeat the calculation if teams are restructured. Do shareholders support the change?

Assume now that "Aerospace 27" is not all-equity financed but holds a long-term debt with a face value of €600 million maturing in 2028. The tax savings from the debt increase the value of the project (in all states) and are equal to 4% of the face value of debt.

- c. Does the manager restructure teams if the company is levered? How high is the agency cost of debt? Show that the expected value of the levered company can be calculated by subtracting the agency cost of debt and adding the benefit of debt to the value of the unlevered company (hint: Adjusted Present Value Method)

Mini Case 4.4

Assume that the risk-free interest rate is 0 and all agents are risk-neutral. Petron Corporation's management team is meeting to decide on a new corporate strategy. There are four options, each with a different probability of success and total firm value in the event of success, as shown below:

	Strategy			
	A	B	C	D
Probability of success	85%	70%	50%	40%
FCF if Successful (in €million)	65	75	50	40

Assume that for each strategy, FCF is zero in the event of failure.

- a Which strategy has the highest expected payoff?
- b Suppose Petron's management team will choose the strategy that leads to the highest expected value of Petron's equity. Which strategy will management choose if Petron currently has:
 - I) No debt?
 - II) Debt with a face value of €15 million?
 - III) Debt with a face value of €30 million?

What agency cost of debt is illustrated in the choice of Petron?

Mini Case 4.5 The value of a share in a company is worth either USD 120, USD 100, or USD 80. Investors think that each case is equally likely and therefore the current share price is USD 100. A large part of the CEO's wealth is invested in the company's stock. The lack of diversification reduces the value of the stock to her by 3%. The CEO announces that she intends to sell half of her shares in the company. Assume that the sale will be executed at the share price that prevails after the share price has adjusted to the news of the sale will be made.

- a Assume that the CEO has no private information about the value of the company's shares. What will be the share price after the CEO announces that she intends to sell half of her shares?
- b Instead of the assumption in a), assume that CEO knows the true value of a share. Investors are rational and know that the CEO knows the true value of a share. What is the share price after the CEO announces that she intends to sell half of her shares?

Mini Case 4.6

A company has two investment opportunities, replacing an old production line with a new one using similar technology (denoted as IO1) and investing in a new production line with risky brand-new technology (denoted as IO2). The cash flows of each investment opportunity at the end of the year are as follows:

	IO1 (old technology)		IO2 (new technology)	
Probability	50%	50%	50%	50%
Payoff (€)	1000	1000	1600	300

Each project requires an initial investment of 500€. The company can only invest in one of the two projects. All players are risk neutral; the risk-free interest rate is 0 and there are no taxes. Assume that management acts in the interest of shareholders.

- Which project yields the highest net present value?
- Which project does the management choose if the company is 100% equity financed?
Calculate the equity value.

The company finances the investment cost of €500 by taking out a loan. However, the lending bank cannot observe which investment opportunity the company chooses after the loan is granted but anticipates the shareholders' optimal decision.

- Which project does the management choose if the project is financed with 500€ debt assuming that the firm needs to pay back 500€ by the end of the period?
- Taking into consideration the possible choice of the management, what is the repayment requirement of the bank for this €500 loan?
- Calculate the value of equity.
- What agency cost is illustrated in your answers above and how high is the expected agency cost?